

# Trustworthy Deep Learning for Pharmaceutical Tablet Quality Prediction: End-to-End Multi-Task Modelling with Dual Uncertainty Quantification on Industrial Compression Trajectories

Project Report

MSc (Research) Practicum  
Artificial Intelligence

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August 2026

# Trustworthy Deep Learning for Pharmaceutical Tablet Quality Prediction: End-to-End Multi-Task Modelling with Dual Uncertainty Quantification on Industrial Compression Trajectories

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**Background.** Every batch a pharmaceutical tablet press runs, that is logged in detail, but most of that sensor data is used just only to check that the line is behaving, and not to stay ahead of time whether the batch will pass its quality specification. The published work tends to boil and filter each run down to just a few summary statistics before any of the modelling starts, which majorly throws away how the process actually unfolds over time, and it kind of hands back a single number with no dependable sense of how much to trust that number.

**Objectives.** The question we have on the table is that whether a model that can read the raw compression trajectory can that model predict four laboratory quality attributes at the same time, whether that one shared encoder across those attributes beats training a separate model for each, and whether the uncertainty that comes out of it is calibrated and explainable enough to stand up in a regulated setting atmosphere.

**Methodology.** The proposal is the use of MT-TrajNet: It is a dilated temporal convolutional encoder with is shared across four of the evidential regression heads, which is trained end to end on 1,005 production batches. This evaluation uses a cross validation grouped by the product code, a distribution shift test with a matched control, corrected resampled paired  $t$ -tests, and an attribution split by the model’s own confidence.

**Results.** While using 3.73 times fewer parameters, the multi task model matches four separate networks, and is better on dissolution significantly. On three of the four attributes, its intervals sits near nominal. The product that is unseen and falls inside the trained hardness range the error rate drops to 0.64 times the cross validation reference, and climbs to 2.43 times on one outside that range. As the model grows less confident, the sensors it leans on shift measurably ( $\tau = 0.822$ ,  $p = 0.0004$ ).

**Findings.** When set against the engineered feature baselines, the model is comparable on two of the attributes and was behind on two as well. The shortfall is kind of structural as the error across the held out codes scales with the distance from the training distribution ( $r = 0.941$ ). The contribution is then a trajectory native MT-model whose uncertainty reacts to a real change of the regime and the explanations of it depends on how confident it is.

**Keywords:** Trustworthy ML, Time series deep-learning, ML learning, uncertainty quantification, pharmaceutical manufacturing

## 1 Introduction

The tablet manufacturing can be considered as one of the most heavily regulated production processes there are. Before they are released for consumption, each and every batch has to pass through a series of laboratory testing for dissolution, mechanical strength,

hardness and weight uniformity, and a failed batch means a regulatory reporting, leading to delays and reworks (Žagar and Mihelič, 2022). The quality by design set out by Yu et al. (2014), asks the manufacturers to tie the critical quality attribute part of a product back to the process parameters that drive them. Modern rotary presses are densely instrumented: they record compression force, ejection force, die fill and other signals every few seconds across the whole run. In practice, though, that stream is used to confirm the machine is running rather than to anticipate whether the product will actually pass.

## 1.1 Background and motivation

Three limitations run through the existing work on predicting tablet quality from process data.

The first is that the process data gets compressed before anyone uses it. Susto and McLoone (2015) predicted slow-release dissolution profiles with LASSO regression on engineered summary features, and Wang et al. (2025) pushed that line further with a graph attention ensemble spanning production units. Both take a run lasting several hours and reduce it to a handful of aggregate numbers. Whatever the shape of the signal over time was carrying is gone before modelling even begins.

The second is that quality attributes are modelled one at a time, despite being physically linked. Hardness, tensile strength, dissolution and weight uniformity all depend on how much material enters the die and how hard it is compressed, so treating them as unrelated regression problems discards structure that is there for the taking.

The third, and the one that matters most in a regulated setting, is that these methods return a number without a trustworthy statement of confidence. Shah et al. (2025) interviewed practitioners in safety-critical industries and found that unexplained model output creates compliance risk and forces manual re-validation, which cancels out much of what automation was supposed to buy. A prediction that cannot flag when it should not be believed is of limited use when the fallback is a laboratory test.

A fourth problem cuts across the other three. Reported improvements in applied machine learning often do not survive appropriate statistical testing (Mahynski et al., 2025, Nadeau and Bengio, 2003), so a claimed gain is not, on its own, evidence of one.

## 1.2 Research question and objectives

These gaps lead to the following research question:

Can an end-to-end multi-task deep learning model operating on raw pharmaceutical compression trajectories produce quality predictions that are competitive with engineered-feature baselines, exploit the correlations between quality attributes, and supply calibrated uncertainty that remains informative when the model is applied to unseen product conditions?

That question splits into four parts. These labels are used throughout the report and match the ones used in the accompanying presentation.

- **RQ1.** Does modelling the raw trajectory beat engineered-feature baselines?
- **RQ2.** Does one shared encoder across the four attributes beat training four separate models?

- **RQ3a.** Is the uncertainty calibrated, and does it stay informative when the model meets an unseen product?
- **RQ3b.** Does the model change which sensors it relies on as its own confidence falls?

To address this question the following objectives were derived:

1. Survey the state of the art in pharmaceutical quality prediction, time-series representation learning, uncertainty quantification and explainability for regulated deployment, and pin down where they fail to meet.
2. Design a multi-task architecture that reads raw compression trajectories and returns, from a single forward pass, a prediction with an uncertainty estimate for each of four quality attributes.
3. Implement the model alongside engineered-feature baselines, single-task and point-output variants, and a deep-ensemble reference for uncertainty.
4. Evaluate accuracy against those baselines using corrected statistical tests, calibration against nominal coverage, behaviour under a genuine distribution shift, and whether the model’s explanations change with its own confidence.

### 1.3 Scope, limitations and assumptions

There are several limitations which are not discussed but stated cleanly upfront.

The published dataset which is used is from a single production line, and therefore nothing can be claimed about how to transfer it to other equipment. In the cross validation three folds is used which are grouped by the product code, leakage is kept out using grouping, but the three folds as a result leave the corrected paired test with very little power, so most of the differences are then reported as inconclusive rather than as equivalence. The hyperparameters were not searched but were fixed. As the dataset, it does not carry any release specification limits, so the work it predicts the continuous quality values and does not classify batches as just a pass or fail.

The work also considers and assumes that the measurements that are provided are true, and that the authors anonymisation preserves the physical relationships in the data, and the product code is a good enough grouping unit to prevent the leakage problem.

### 1.4 Contribution

MT-TrajNet is the main contribution as a trajectory native ML-model with dual uncertainty output, and is assessed properly about where this model properly works and what are its shortcomings. There are three of the major outcomes. First is an architecture that matches four separate networks at 3.73 times fewer parameters, Second is a demonstration that the uncertainty and error they both react to a genuine change of regime while staying stable on an unseen product inside the trained range, Third is a significant result that the model, it reshuffles which of the sensors it depends on as when the confidence falls, homing in on the signals that physically govern the attribute.

### 1.5 Report Structure

In Section 2, it reviews the prior work done across the four areas and then it positions and is put against this study. In Section 3 we describe the data, discover and shape

the evaluation design and the experimental protocol of it. In Section 4 we specify the architecture. In section Section 5 we cover what was built. Section 6 it reports four experiments and discusses about it and how they imply, and in the Section 7 we discuss what we were able to achieve and what comes next.

## 2 Related Work

This study is basically drawn on four bodies of work, first being ML for pharmaceutical quality prediction, second being the uncertainty quantification, third being the representation learning for industrial time series, and lastly explainability for regulated deployment. Each of them are critically reviewed below, and Table 1 sets the closet previous studies against the present work.

### 2.1 Machine learning for pharmaceutical quality prediction

Susto and McLoone (2015), in *Computers in Biology and Medicine*, is the study of reference for this issue. In this they predict the slow release dissolution profiles from the granulation and coating data using the Lasso regression over the engineered features with Monte Carlo cross validation and then showed that dissolution is in fact predictable from the process data and that the sparse linear models, they hold their own on small samples. Its strength is a careful validation protocol on a genuinely industrial problem. Its limitation is exactly the one this study targets: the compression trajectory is reduced to summary statistics before modelling, a single attribute is predicted, and no uncertainty is reported.

Žagar and Mihelič (2022) released the dataset used here, in *Scientific Data*. The contribution was the data collection itself, covering 1,005 batches across 25 product codes, with benchmark predictions from engineered features. Its strength is scale and openness, which is what makes independent comparison possible at all. Its limitation is that the raw trajectories they published are never modelled directly in their own benchmarks, which leaves the central question of this study wide open.

The most sophisticated recent work is Wang et al. (2025) in *Acta Pharmaceutica Sinica B*: end-to-end quality prediction across six production units, using a path-enhanced double-ensemble graph attention network, with prediction improved by modelling inter-unit dependencies. Its strength is that it captures structure that flat feature models cannot. Its limitations are that it still runs on aggregate engineered features, uses a small sample, and reports neither uncertainty quantification nor evaluation under distribution shift.

Aghajanpour et al. (2025) review machine learning for polymeric drug delivery and make a point that lands directly here: neural networks often beat classical models on small benchmarks, but those comparisons frequently skip statistical validation and the models are rarely interpretable. Together with Mahynski et al. (2025), who show that apparent gains often fall apart under correct testing, this is the reason the corrected resampled paired  $t$ -test of Nadeau and Bengio (2003) is used throughout.

Taken together, these studies share a pattern: each treats a manufacturing run as a bag of numerical features rather than a signal that evolves over time, none benchmarks a trajectory model against the engineered-feature state of the art, and none supplies calibrated uncertainty.

## 2.2 Representation learning for industrial time series

Outside the pharmaceutical domain, sequence modelling has moved on a good deal. Bai et al. (2018) showed that temporal convolutional networks, built from stacks of dilated causal convolutions, match or beat recurrent architectures across a wide range of sequence benchmarks while training more stably and in parallel. The strength of that result is that it makes long-sequence modelling practical without recurrence; its limitation for present purposes is that it is shown on generic benchmarks rather than on industrial sensor data.

Nie et al. (2023) introduced channel-independent patching for long-horizon forecasting with transformers, which is efficient and strong on forecasting tasks, but attention cost still grows with the square of the sequence length. That is a real constraint here, because runs in this dataset stretch past forty thousand samples before downsampling, and it is why a dilated convolutional encoder was preferred (Section 4). The patch-based alternative is an ablation scoped out of this study rather than one tested and rejected.

## 2.3 Uncertainty quantification

Treating a prediction as a distribution rather than a point estimate goes back a long way in the probabilistic view of machine learning (Murphy, 2012), but two families dominate its practical use in deep regression. Amini et al. (2020) proposed deep evidential regression, which swaps the point output for the parameters of a Normal-Inverse-Gamma distribution and so returns aleatoric and epistemic uncertainty from a single forward pass. Its strength is efficiency; its known weakness, acknowledged by the authors and confirmed independently, is that the epistemic estimate can be poorly calibrated once the data drifts away from training conditions. Lakshminarayanan et al. (2017) showed that deep ensembles give well-calibrated uncertainty from the disagreement between independently trained networks, at a cost that scales with ensemble size. Conformal prediction (Shafer and Vovk, 2008) which is the third route, wraps any of the predictor in the intervals that carry a coverage guarantee based on the assumption that the data is exchangeable.

Hüllermeier and Waegeman (2021) give the conceptual split between aleatoric and epistemic uncertainty that this study relies on, and Guo et al. (2017) showed that modern networks are systematically overconfident and that post-hoc calibration is usually necessary rather than optional. Specifically for the regression part of it, Kuleshov et al. (2018) to hit there normal coverage predicted intervals can be recalibrated, which is what the principle is behind the scale correction applied in Section 3.6.

Closest to the central claim of this whole study is Ovadia et al. (2019) which evaluated and studied the predictive uncertainty under the dataset shift across several methods and found that the calibration and accuracy they both degrade as the shift grows, with ensembles degrading the most very gracefully. Synthetic corruptions of standard benchmarks were used in their evaluation. Through this what that leaves open is whether the same conclusions do they hold on to real industrial data where the shift arises naturally from a change of product rather than from an artificial perturbation, what is what is there in the Section 3.5.

## 2.4 Explainability for regulated deployment

Interpretability is a basic requirement in high stakes settings and is not just an optional extra which is what is told by Rudin et al. (2022). Lundberg and Lee (2017) introduced SHAP, which unifies several earlier attribution methods and is the dominant post-hoc

technique in practice; Sundararajan et al. (2017) provide integrated gradients as the main axiomatic alternative. For this application both share a limitation: every prediction is treated as equally explainable, and attribution is computed without any reference to how confident the model is.

Shah et al. (2025) found that non-explainable output creates compliance risk and extra manual validation in regulated industries. Read next to Rudin et al. (2022), this points to a stronger requirement than attribution on its own: a confident prediction and an uncertain one should not be explained the same way, because an operator would act on them differently. Closing that gap is the aim of Section 6.4.

## 2.5 Positioning and research gap

Table 1 compares the closest prior studies against this work. The comparison is deliberately drawn on method and capability rather than on reported error values, because the studies use different datasets, different quality attributes and different evaluation splits, which would make a direct numerical comparison misleading.

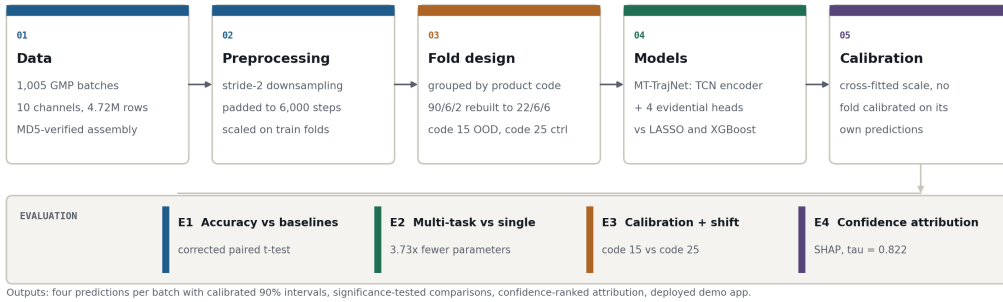
**Table 1:** Capability comparison with the closest prior work. UQ denotes calibrated uncertainty quantification; OOD denotes an explicit test under distribution shift.

| Study                     |  | Data                         | Input                      | Targets         | UQ  | OOD | Significance testing  |
|---------------------------|--|------------------------------|----------------------------|-----------------|-----|-----|-----------------------|
| Susto and McLoone (2015)  |  | Tablet manufacturing         | Engineered features        | 1 (dissolution) | No  | No  | Monte Carlo CV        |
| Žagar and Mihelič (2022)  |  | 1,005 GMP batches            | Engineered features        | Several         | No  | No  | Benchmark only        |
| Wang et al. (2025)        |  | Six-unit plant, small sample | Engineered features, graph | Several         | No  | No  | Not corrected         |
| Aghajanpour et al. (2025) |  | Review                       | Mixed                      | Varies          | No  | No  | Noted as often absent |
| <b>This work</b>          |  | 1,005 GMP batches            | Raw 10-channel trajectory  | 4, jointly      | Yes | Yes | Nadeau–Bengio         |

In short: the pharmaceutical literature works from engineered features and never models the raw trajectory; the sequence-modelling literature supplies suitable architectures but does not evaluate them for calibration or interpretability in a regulated context; the uncertainty literature produces methods rarely tested on industrial data under naturally occurring shift; and the explainability literature does not condition explanations on predictive confidence. No existing study brings these together, and that combination is what this research addresses, feasibly on the Žagar and Mihelič (2022) dataset without any additional laboratory work.

### 3 Research Methodology

This study follows an empirical, experimental research strategy in the sense of Saunders et al. (2023). The goal is to predict four laboratory quality attributes of pharmaceutical tablets straight from the sensor trajectories recorded during compression, and then to test how well the model can judge its own uncertainty. From the raw sensor data to the predictions evaluated is provided in this section. The steps follow the data-understanding, preparation, modelling and evaluation stages of CRISP-DM, but throughout the emphasis is on a reproducible experimental protocol rather than a fixed process template. The Figure 1, it provides the whole pipeline structure that the rest of this section then walks through step by step.



**Figure 1:** Shows the full research pipeline end to end from the raw sensor files through preprocessing, fold design and model training to calibration and the four of the evaluation experiments.

#### 3.1 Dataset

The study uses the public tablet-compression dataset of Žagar and Mihelič (2022), published in *Scientific Data* under a CC BY 4.0 licence. It contains 1,005 production batches of a film-coated tablet, spanning 25 product codes with several strength variants. Each batch is a multivariate time series from a rotary tablet press, ten sensor channels sampled every ten seconds, which comes to roughly 4.72 million sensor rows in total. The ten channels are tablet press speed, filling device speed, main compression force, tablet fill, the slow-release indicator (SREL), pre-compression force, cylinder main compression height, cylinder pre-compression height, stiffness and ejection force.

Trajectory length varies a lot, because batch sizes and run times differ: the shortest run has 993 samples, the median 3,228, and the longest 41,710. Each batch carries four laboratory-measured targets: average dissolution (82.50 to 102.67, mean 90.65), average hardness (27.00 to 102.00, mean 54.87), weight relative standard deviation (0.41 to 9.90, mean 1.09) and tensile strength (1.04 to 3.04, mean 1.67). The targets are only moderately correlated, for instance dissolution and tensile at 0.52 and hardness and tensile at  $-0.38$ , while weight uniformity is close to independent of the rest. That partial overlap is the regime where a shared multi-task model can help: there is common structure to exploit without the tasks being redundant.

For reproducibility, all the trajectories are assembled into a single serialised file whose MD5 fingerprint (7d7fc76be1e4940198b76d9d0797a3a9) is asserted at the start of every modelling notebook, so any change in the data is caught straight away.

### 3.2 Bimodal hardness and product-code structure

An early look at the targets showed that hardness is not spread evenly; its distribution is bimodal. A total of 98 batches sit above 70 N in a high-hardness cluster, and the remaining 907 sit at or below 70 N. Tracing those 98 batches back to their product codes, they map cleanly onto just five codes (4, 8, 15, 20 and 24), with no overlap into the low cluster, and are split very much unevenly among five of those, where the code 15 alone it holds 64, the code 4 holds 22 and the code 24, 20 and 8 they hold 6, 4 and 2 respectively.

This one thing actually shapes the rest of the design as the high hardness regime is concentrated in a handful of code and is dominated by one of them in particular, so any evaluation split that groups data by the product code risk pushing almost the entirety into one single fold.

### 3.3 Data preparation

Each of the trajectory is downsampled and it put to a two strides, which roughly halves its length while keeping the shape of every signal. All the ten of the sensors are kept and then each channel is standardised to zero mean and the unit variance. The mean and standard deviation are computed only on the training batches of the fold which are then applied without any change to the validation and the test batches, so this results in scaling never knowing the held out data. The four targets are also standardised the same way during the training and then are mapped back to their physical units for reporting. After they are standardised each of the sequence is then zero padded or then truncated to a fixed length of 6,000 steps, so that batches of unequal length can be also processed together.

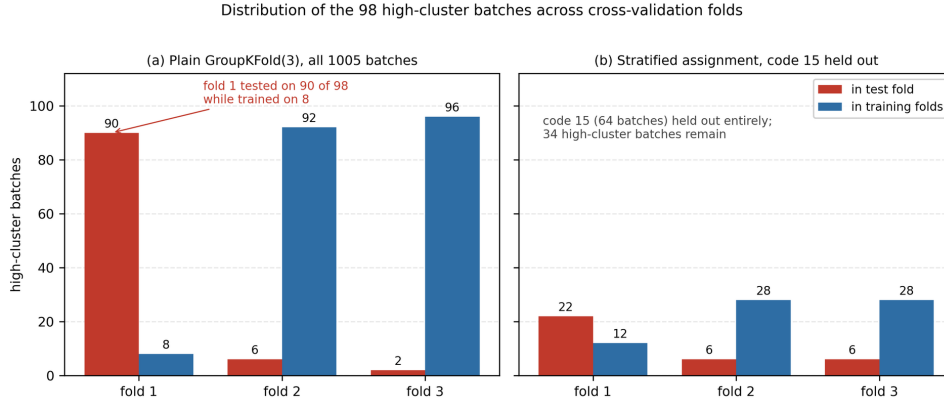
### 3.4 Cross-validation design

The cross validation is grouped on the basis of the product code, making sure that no code appears on both the test side of a split and the training as well. This helps in blocking the leakage and gives a realistic read on how the model is performing and behaving on products it has never seen. This grouped resampling is the standard remedy when the observations are not independent (Hastie et al., 2009).

The first attempt was a plain three fold GroupKFold and it failed for a very clear reason. It placed 90 of the 98 high-hardness batches into a single test fold, leaving the other two folds with only 6 and 2. That fold was then tested on 90 high-hardness batches while the model had trained on just 8 of them, which inflated the hardness error and made the early numbers unreliable. A diagnostic traced this straight to the fold construction, before anything was changed in the model.

The fix has two parts. First, code 15 is held out of the cross-validation entirely, because at 64 batches it is too large to spread across three folds without dominating one of them. Second, the four remaining high-hardness codes are balanced by hand across the folds to give roughly 22, 6 and 6 high-cluster batches per test fold. The final assignment is that fold 1 holds codes 1, 2, 4, 6, 7, 9, 13, 18 and 25, fold 2 holds codes 10, 11, 17, 19, 22 and 24, and fold 3 holds codes 3, 5, 8, 12, 14, 16, 20, 21 and 23. This leaves 941 batches inside the three-fold cross-validation with code 15's 64 batches reserved. Figure 2 contrasts the broken split with the corrected one.

One consequence is worth stating plainly. Once code 15 is removed, only 34 of the high-hardness batches remain for training across all folds, so hardness stays structurally



**Figure 2:** High-cluster batches per fold under plain GroupKFold (left) and under the cluster-balanced assignment with code 15 held out (right).

hard to predict. This is reported as a limitation rather than engineered around, and Section 6.5 shows that it explains the hardness result.

### 3.5 Out-of-distribution test

Code 15 serves as an out-of-distribution probe. It is a high-hardness code with 64 batches, and no fold ever trains on it. To separate a real change of regime from ordinary unseen product variation, a control is added: code 25, a low-hardness code with 34 batches that the fold 1 model also never sees. The fold 1 model scores both codes, and as a robustness check code 15 is additionally scored by the fold 2 and fold 3 models. No training happens at this stage; saved weights are loaded in inference mode and the uncertainty scale used is the cross-fitted one described below, so it never touched either code. Because the folds are grouped by code, the out-of-fold predictions for several low-cluster codes are themselves leave-one-code-out results at no extra cost.

### 3.6 Uncertainty calibration

A single forward pass returns the parameters of a predictive distribution, from which a mean and a total predictive variance, combining model and data uncertainty, are taken. The raw intervals need a scale correction to be trustworthy, following the principle of recalibrating regression intervals to their nominal coverage (Kuleshov et al., 2018). The correction is done by cross-fitting: each fold’s scale factor is fitted only on the out-of-fold predictions of the other two folds and then applied to that fold, so no batch is ever calibrated on its own prediction. An earlier version fitted the scale on the same predictions it then reported on, which is optimistic and not sound; it was corrected to the cross-fitted version, and in practice the calibrated coverage barely moved. Both the fix and the fact that it changed little are reported here.

### 3.7 Interpretability protocol

To see which sensors the model actually relies on, SHAP is used with a gradient-based explainer (Lundberg and Lee, 2017), 32 background samples, on the full fold 1 test set of 314 batches. For each target, the test batches are split into the most confident 20% and the least confident 20% by the model’s own epistemic uncertainty. The per-channel attributions are computed in each group and the two channel rankings are then compared

with Kendall’s tau. This asks a precise question: does the model change which sensors it depends on when it becomes uncertain? SHAP is chosen over integrated gradients (Sundararajan et al., 2017) because it is efficient at this scale and works cleanly with the evidential output heads.

### 3.8 Evaluation metrics and statistical testing

Point accuracy is measured with RMSE as the primary metric, with MAE and  $R^2$  alongside it, reported per target and both pooled and per fold. Uncertainty quality is measured with prediction-interval coverage at the 80, 90 and 95 nominal levels, with the expected calibration error, and with the correlation between the actual error and the predicted uncertainty. The number of bins is always stated when reporting the calibration error, since it is sensitive to that choice, most visibly for weight RSD. The attribution shift is measured with Kendall’s tau, as above. As a reference point for the evidential uncertainty, the model is also benchmarked against a deep ensemble (Lakshminarayanan et al., 2017).

For the model comparisons, such as multi-task against single-task and the proposed model against the baselines, the corrected resampled paired  $t$ -test of Nadeau and Bengio (2003) is used, which accounts for the dependence introduced by overlapping training folds. With only three folds this test has low power, so most of the differences do not reach significance. Non-significant results are treated as such, and the low power is flagged as a limitation, rather than reading meaning into small gaps.

## 4 Design Specification

MT-TrajNet is a single network with one shared encoder and four prediction heads. It reads the full compression trajectory and returns, in one forward pass, a prediction and an uncertainty estimate for each of the four quality targets. The design aims to model the raw trajectory directly, share structure across the four correlated targets, and produce usable uncertainty without any test-time sampling.

### 4.1 Shared encoder

The encoder is a temporal convolutional network (Bai et al., 2018). It stacks four residual blocks with dilation factors of 1, 2, 4 and 8, where each block uses two causal convolutions of kernel width three, ReLU activation and dropout of 0.1. All four blocks use 128 channels, and a  $1 \times 1$  convolution matches the channel width on the residual path of the first block. Doubling the dilation at each block widens the receptive field quickly, so a block can cover long stretches of a 6,000-step trajectory without any recurrent loop; this is the standard way to enlarge receptive fields without extra depth or pooling (Goodfellow et al., 2016). A global average pooling over the time dimension then collapses the variable-length sequence into a fixed 128-dimensional representation.

A standard Transformer encoder was avoided because its attention cost grows with the square of the sequence length, which becomes too expensive at the long end of this dataset, whereas the dilated stack with stride-based downsampling keeps compute linear in sequence length. The patch-based alternative of Nie et al. (2023) would cut that cost, and comparing the two representations is identified as future work in Section 7.1.

## 4.2 Evidential prediction heads

The pooled 128-dimensional representation feeds four small heads, one per target. Each head is a two-layer perceptron with a hidden width of 64 that outputs the four parameters of a Normal-Inverse-Gamma distribution, following the deep evidential regression formulation of Amini et al. (2020). The Normal-Inverse-Gamma is the conjugate prior for a Gaussian with unknown mean and variance (Gelman et al., 2013), which is what lets a single set of outputs describe both the prediction and the confidence in it. In one pass this gives a mean prediction together with separate epistemic (model) and aleatoric (data) uncertainty (Hüllermeier and Waegeman, 2021), with no sampling and no ensembling at inference time. Training uses the evidential negative log likelihood with the evidence regulariser of the same formulation, weighted at 1.0, which penalises confident evidence on wrong predictions. The parameters  $\nu$ ,  $\alpha$  and  $\beta$  are clamped to  $[10^{-2}, 10^3]$  for numerical stability.

## 4.3 Multi-task loss

The learning related tasks through just one shared representation, it goes back to Caruana (1997), whose argument was that the training signal of each of the task, it acts as an inductive bias on others. The overall training loss is the sum of the four per-target evidential losses, each given equal weight. The four targets are standardised to zero mean and unit variance during training, as described in Section 3.3, which puts them on a comparable scale, so an equal sum is enough to balance them and no per-task weighting is needed. The four losses are optimised jointly through the shared encoder, which is what couples the tasks and lets them share representation.

## 4.4 Parameter efficiency

Sharing one encoder across four light heads keeps the model small. MT-TrajNet has 384,400 parameters. Four separate single-task models of the same design would use 1,435,408 parameters in total, so the shared model is 3.73 times smaller. This is a property of the design itself; the matching accuracy comparison is reported in Section 6.2.

# 5 Implementation

This section describes the final state of what was built. Environment details, library versions and reproduction instructions are given in the accompanying Configuration Manual rather than here.

## 5.1 What was built

The **data pipeline** assembles 25 per-product sensor files into a single serialised array of 1,005 trajectories, joins them to the laboratory measurements by batch identifier, and writes an MD5 fingerprint that every downstream stage asserts. The join matched 1,005 to 1,005 with no orphans. **MT-TrajNet** is the model of Section 4: a shared encoder with four evidential heads, 384,400 parameters, trained with Adam (Kingma and Ba, 2015) at learning rate  $5 \times 10^{-4}$ , batch size 16, up to 150 epochs with early stopping after 10 without improvement, gradients clipped at norm 1.0, seed 42.

**Engineered-feature baselines** were fitted on the 28 batch-level features supplied

with the dataset: a mean predictor, LASSO following Susto and McLoone (2015), and gradient boosting, each in default and tuned configurations, with tuning kept inside the training folds. Three variants isolate design decisions: a single-task variant separating the effect of sharing from that of trajectory modelling, a point-output variant separating the cost of the evidential objective from that of the encoder, and a five-seed deep ensemble as a reference uncertainty estimate.

The **analysis stage** produces out-of-fold predictions and uncertainties for all 941 batches, applies the cross-fitted scale correction of Section 3.6, computes coverage and calibration error, runs the corrected paired tests, and computes the SHAP attributions of Section 3.7. The **demonstration application** presents the study interactively, reading precomputed results rather than running the model, so it needs no GPU.

## 5.2 Outputs and tools

These stages produce an assembled dataset with an integrity fingerprint, trained weights per fold, per-batch out-of-fold predictions and calibrated uncertainties, result files holding every metric reported here, the figures, and the deployed application. The implementation used Python with PyTorch, scikit-learn and XGBoost for the baselines (Géron, 2019), and SHAP for attribution; training ran on a single NVIDIA Tesla T4 and the application uses Streamlit. The notebooks, results and the code they are available publicly but the model weights were excluded because of their size. Each and every experiment it runs from the same fixed seed and the file so that the numbers that are reported can be reproduced. I attached the link to my public Github repository for this project as well (<https://github.com/arpitelias/mt-trajnet-thesis>). The Application is also deployed publicly at (<https://mt-trajnet-thesis-4kznhd8sjhv28lcfds4e2k.streamlit.app/>).

## 6 Evaluation

Four experiments are reported: accuracy against engineered-feature baselines, the effect of sharing one encoder across four targets, calibration and behaviour under a genuine change of operating regime, and whether the model’s explanations depend on its own confidence. All four use the same folds, seed and data described in Section 3, so the comparisons are like for like. Negative results are reported next to positive ones.

### 6.1 Experiment 1 (RQ1): accuracy against engineered-feature baselines

Table 2 reports out-of-fold RMSE for every model on all four targets. Lower is better, and the best value in each row is shown in bold.

**Table 2:** Out-of-fold RMSE by model and target. Units follow the target: percent for dissolution and weight RSD, newtons for hardness, megapascals for tensile strength.

| Target           | Mean predictor | LASSO        | LASSO (tuned) | XGBoost      | XGBoost (tuned) | MT-TrajNet |
|------------------|----------------|--------------|---------------|--------------|-----------------|------------|
| Dissolution      | 3.533          | <b>3.171</b> | 3.205         | 3.272        | 3.243           | 3.261      |
| Hardness         | 8.820          | 5.962        | 6.087         | <b>4.737</b> | 4.761           | 7.698      |
| Weight RSD       | 0.573          | <b>0.557</b> | 0.563         | 0.635        | 0.611           | 0.585      |
| Tensile strength | 0.422          | 0.359        | 0.325         | 0.279        | <b>0.270</b>    | 0.345      |

MT-TrajNet clears the mean predictor on every target, which confirms that the raw

trajectory carries usable signal for all four attributes. Against the strongest engineered-feature baseline it is within roughly three percent on dissolution and five percent on weight uniformity, about twenty-eight percent behind on tensile strength, and about sixty-three percent behind on hardness.

The corrected resampled paired  $t$ -test of Nadeau and Bengio (2003) was applied against the tuned gradient boosting baseline. Only the hardness difference reaches significance ( $p = 0.015$ ). Dissolution ( $p = 0.886$ ), weight uniformity ( $p = 0.697$ ) and tensile strength ( $p = 0.269$ ) do not. With three folds this test has very little power, so the three non-significant results should be read as inconclusive, not as demonstrated parity; the tensile gap in particular is numerically substantial even though the test cannot resolve it.

The answer to RQ1 is therefore partial, and reported that way: modelling the raw trajectory is competitive on two attributes and behind on two, and does not deliver a uniformly more accurate predictor. Section 6.5 shows the hardness result has a structural cause in the dataset rather than in the model.

A third variant separates the cost of the evidential objective from that of the encoder. With an identical encoder and folds but a linear output under mean squared error, the point-output model reaches 6.698 N on hardness against 7.698 N, so the evidential objective costs about one newton. It still trails tuned gradient boosting by roughly 1.9 N, so most of the hardness gap comes from the encoder and the available training data rather than the evidential formulation. On the other three targets it is equal or slightly worse, so the uncertainty costs very little accuracy elsewhere.

## 6.2 Experiment 2 (RQ2): one shared model against four separate ones

The single-task comparison keeps the architecture identical and removes only the sharing. Four networks were trained, one per target, on the same folds.

The multi-task model is better on all four targets: 3.261 against 3.547 on dissolution, 7.698 against 8.310 on hardness, 0.585 against 0.595 on weight uniformity, and 0.345 against 0.364 on tensile strength. The dissolution improvement is statistically significant ( $p = 0.012$ ); the other three fall within the noise of three folds.

It reaches this using 384,400 parameters against 1,435,408 for four separate networks, a reduction of 3.73 times. So sharing a trajectory encoder across correlated attributes does not cost accuracy and cuts model size substantially, which matters in a setting where four separate models would otherwise be maintained, validated and version controlled on their own.

## 6.3 Experiment 3 (RQ3a): calibration and behaviour under distribution shift

Table 3 reports interval coverage at the 90 percent nominal level, expected calibration error using ten bins, and the correlation between predicted uncertainty and actual error.

Reading the three coverage columns together says more than any single one. Dissolution climbs from 0.770 to 0.897 as the nominal level moves from 80 to 95 percent, and hardness from 0.846 to 0.957: in both, the interval responds to the level requested, which is what calibration means. Weight uniformity does not, moving only from 0.930 to 0.960, so asking for a wider interval barely produces one. That flatness, not the absolute

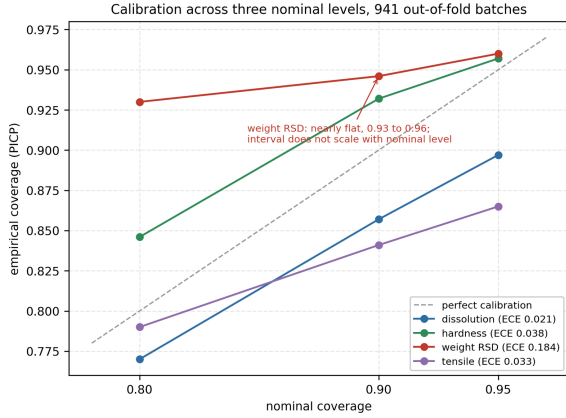
**Table 3:** Uncertainty quality on the 941 out-of-fold batches. Coverage should track the nominal level; calibration error and interval width relative to error should be low; a higher uncertainty–error correlation means the uncertainty is more informative about which batch is wrong.

| Target           | Empirical coverage |       |       | Calib. error<br>(10-bin) | Interval width<br>÷ RMSE | Unc.–error<br>correlation |
|------------------|--------------------|-------|-------|--------------------------|--------------------------|---------------------------|
|                  | 80%                | 90%   | 95%   |                          |                          |                           |
| Dissolution      | 0.770              | 0.857 | 0.897 | 0.021                    | 3.32                     | +0.035                    |
| Hardness         | 0.846              | 0.932 | 0.957 | 0.038                    | 3.30                     | +0.533                    |
| Weight RSD       | 0.930              | 0.946 | 0.960 | 0.184                    | 3.26                     | +0.029                    |
| Tensile strength | 0.790              | 0.841 | 0.865 | 0.033                    | 3.40                     | −0.015                    |

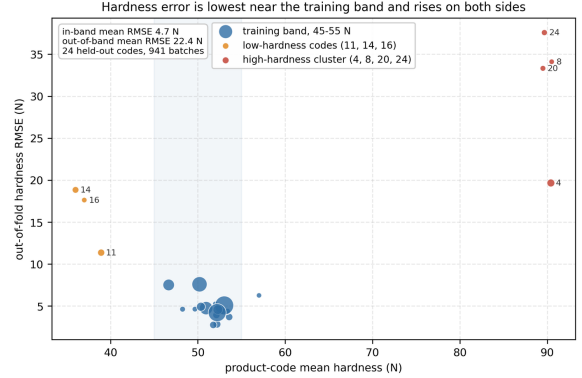
coverage value, is the real defect, and it is why its calibration error is roughly five times that of the other targets. Its intervals should not be relied on for an individual batch.

Interval width relative to the error it must cover sits between 3.26 and 3.40 for all four targets, close to what a correctly scaled Gaussian gives, so the coverage above is achieved by calibration rather than by making the intervals uselessly wide. Hardness is the one target where the uncertainty carries real information about the error (+0.533); on the other three the intervals are well sized on average but say little about which particular batch is likely to be wrong. An average that is correct is not the same as a per-batch warning that is useful.

Figure 3a shows the same behaviour across all three nominal levels. Dissolution, hardness and tensile strength track the diagonal reasonably closely, while weight uniformity is almost flat: its coverage barely moves between the 80 and 95 percent levels, which is what a miscalibrated interval looks like.



(a) Empirical against nominal coverage at three levels. A diagonal is followed by a well calibrated model.



(b) Out-of-fold hardness error for each held-out product code against that code's mean hardness. Error is lowest near the centre of the training distribution and rises on both sides.

Two product codes the fold 1 model was never trained on is compared in the distribution test shift. Code 25 sits inside the hardness range the model learned; code 15 sits in the high-hardness regime it barely saw. Against a cross-validation reference of 7.698 N, hardness error on code 25 falls to 4.918 N, which is 0.64 times the reference, while on code 15 it rises to 18.69 N, which is 2.43 times the reference. Scoring code 15 with the fold 2 and fold 3 models gives the same picture.

That contrast is the most informative result in this section. An unfamiliar product is not, by itself, a problem for the model; an unfamiliar operating regime is. The model’s own uncertainty follows the same split, with median calibrated hardness uncertainty roughly three and a half times higher on code 15 than on batches from the trained regime, so a batch the model cannot judge is visible as such before its prediction is acted on. This is the behaviour Ovadia et al. (2019) found on synthetic corruptions, seen here on a shift that arises naturally from a change of product.

The five-seed deep ensemble was evaluated on the same shifted code as a reference. It is more accurate on all four targets under shift, most clearly on hardness at 14.08 N against 18.69 N, about a quarter lower error. The two are comparable in distribution. The fair conclusion can also be considered as a not a clear winner but just as a trade off as the single pass evidential model, it provides most of the benefit at roughly one-fifth of the training and inference cost, as on the other hand the ensemble is the more robust choice where shift is expected and the compute is available as well. This comparison was run on a single fold and therefore should be read as directional. A change of product code breaks is exactly the exchangeability that the conformal coverage rests on, and Tibshirani et al. (2019) show that getting those guarantees back needs the shift to be known and weighted for.

## 6.4 Experiment 4 (RQ3b): uncertainty-stratified attribution

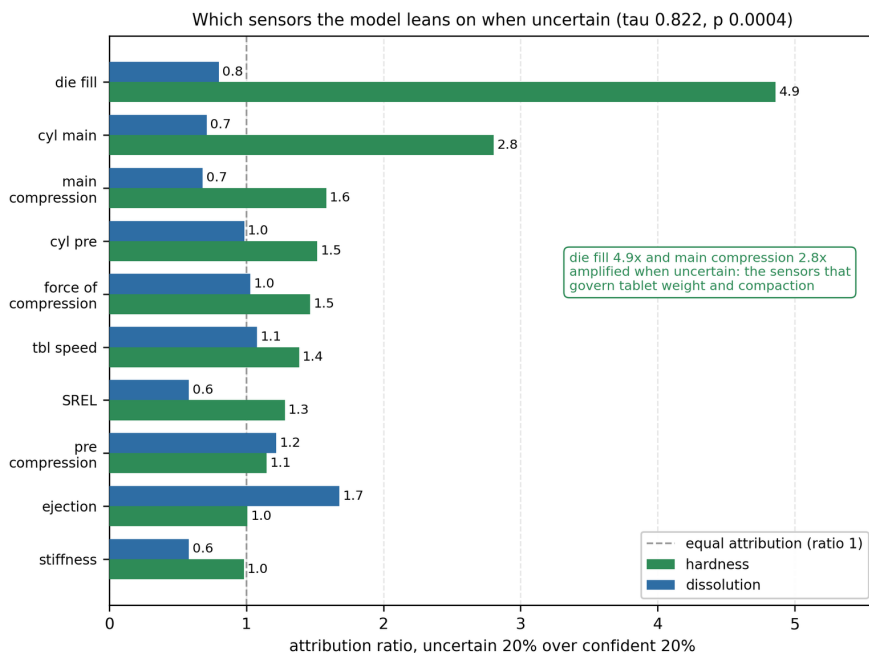
The final experiment asks whether the model’s explanation of a prediction depends on how confident it is. For each target the held-out batches were split into the most confident and least confident fifths by the model’s own epistemic uncertainty, and the two per-channel attribution rankings were compared with Kendall’s tau.

The rankings shift significantly on dissolution and hardness ( $\tau = 0.822$ ,  $p = 0.0004$ ). They do not shift significantly on weight uniformity or tensile strength ( $\tau = 0.467$ ,  $p = 0.073$ ), and both outcomes are reported. Only the first two support the claim.

Figure 4 shows the effect on hardness. When the model is least confident, its dependence on die fill rises by a factor of 4.86 and on cylinder main compression by a factor of 2.81, relative to the batches it is most confident about.

The two channels are those signals that govern how much of the material enters the die and how hard it should be compressed, which in fact what physically determines tablet hardness. The models was also handed 10 raw sensor channels and 4 lab measurements and was never actually told which channels were actually mechanically relevant. The evidence that the representation it learned reflects the process rather than an arbitrary correlation is that its uncertainty concentrates on them and it is therefore the strongest statistical result in this study.

With positives there are negatives as well. Earlier version of this analysis which was run before the cross validation folds were corrected and indicated a large shift in *when* along the path that the model attended and not only in which of the channels. The same phase level effect did not reappear once the folds were stratified properly across two run which ran separately. it was an artefact of the imbalanced folds rather than a property of the model, and is excluded from the findings. The channel level results reported above survived the exact correction and became even stronger.



**Figure 4:** Attribution ratio between the least confident and most confident fifths of held-out batches. A value above one means the model relies on that channel more heavily when it is uncertain.

## 6.5 Discussion

### Confidence in the results

The comparisons are internally consistent: identical folds, a fixed seed, one fingerprinted data file, and a paired test that accounts for dependence between overlapping training sets. The principal threat is statistical power, and Table 4 makes the reason concrete rather than leaving it assumed.

**Table 4:** MT-TrajNet RMSE for each fold. The spread column is the largest fold divided by the smallest, and shows how much of the observed variation is attributable to which product codes happen to fall in a fold.

| Target           | Fold 1 | Fold 2 | Fold 3 | Mean  | Spread |
|------------------|--------|--------|--------|-------|--------|
| Dissolution      | 3.344  | 3.147  | 3.292  | 3.261 | 1.06×  |
| Hardness         | 7.900  | 7.428  | 7.767  | 7.698 | 1.06×  |
| Weight RSD       | 0.770  | 0.442  | 0.543  | 0.585 | 1.74×  |
| Tensile strength | 0.404  | 0.350  | 0.283  | 0.345 | 1.43×  |

Dissolution and hardness vary by only about six percent across folds, which is why the hardness comparison was able to reach significance. Weight uniformity and tensile strength do not: weight RSD varies by a factor of 1.74 between folds and tensile by 1.43. For those targets the variation caused by which product codes land in a fold is of the same order as the differences being tested between models, so no test on three folds could separate them. That is a property of the data and the grouping requirement rather than of the models, and it is the direct reason every non-significant result here is described

as inconclusive rather than as equivalence. Pooled metrics agree: hardness has an  $R^2$  of 0.252 with a 95 percent confidence interval of 0.126 to 0.349.

### **Why hardness behaves differently**

Rather than leave the hardness result unexplained, the error was examined across every product code held out in turn. Figure 3b plots hardness error for each of the 24 codes against that code’s mean hardness.

The relationship is strong and regular: error grows with distance from the centre of the training hardness distribution, with a correlation of  $r = 0.941$  across the 24 codes. The model is not failing at hardness in general. It is accurate near the centre and degrades predictably towards the extremes, where it has seen very few examples.

This has a structural cause established in Section 3.2. The high-hardness regime lives in five product codes, one of which holds 64 of the 98 batches. Because evaluation groups whole codes to prevent leakage, that regime cannot be spread evenly across folds, and once the dominant code is withheld for the distribution-shift test only 34 high-hardness batches remain for training. Gradient boosting on engineered features is less exposed to this because it does not have to learn a representation of the signal from scratch, which is exactly the advantage of a lower-capacity model when data in a region is scarce.

### **Relation to previous work**

The results agree with Aghajanpour et al. (2025) that neural models do not automatically outperform classical methods on modest industrial datasets, and with Mahynski et al. (2025) and Nadeau and Bengio (2003) that apparent gains require corrected testing. They agree with Ovadia et al. (2019) that performance degrades under shift and that ensembles degrade more gracefully, extending that from synthetic corruptions to a natural change of product. They part with prior pharmaceutical work in one respect: Susto and McLoone (2015) and Wang et al. (2025) report point predictions from engineered features, whereas this study reaches comparable accuracy on two of four attributes without feature engineering, and adds calibrated uncertainty and confidence-dependent explanation.

### **Scope, generalisability and limitations**

The scope is one production line, one press and one dataset, so the learned representation may be specific to this equipment, and cross-site validation would be needed before claiming otherwise. The attribution finding is significant on two of four targets and should not be generalised to the other two. Weight uniformity is poorly calibrated and its intervals are not dependable at batch level. Hyperparameters were fixed rather than searched, so the reported accuracy is a lower bound. The evidential-against-ensemble comparison was run on a single fold.

The strength of the approach is that it removes the feature-engineering step, predicts four attributes from one model, and supplies uncertainty and confidence-conditioned explanations that feature-based baselines do not offer. Its weakness is that it does not yet beat a well-tuned gradient boosting baseline on every attribute, and on hardness it clearly does not.

### **Practical implications**

Academically the contribution is the combination: a trajectory-native multi-task model whose uncertainty responds to real regime shift and whose explanations are conditioned

on its own confidence, evaluated with corrected statistics on industrial data. For a practitioner the useful output is not a replacement for laboratory testing, which this work does not claim, but a triage signal: a model that flags an unfamiliar regime by widening its own intervals, and points to the mechanically relevant sensors when it does so, can sit alongside the existing quality process. The hardness relationship reads practically too, in that the most valuable batches to collect are those furthest from the centre of the current distribution.

## 7 Conclusion and Future Work

This research asked whether an end-to-end multi-task model operating on raw compression trajectories could produce quality predictions competitive with engineered-feature baselines, exploit the correlations between quality attributes, and supply calibrated uncertainty that stays informative on unseen product conditions.

All four objectives were met: the state of the art was reviewed across four areas and the gap identified; MT-TrajNet was designed and implemented; baselines and single-task, point-output and deep-ensemble variants were built alongside it; and all were evaluated with corrected statistical tests, calibration analysis, a distribution-shift test with a matched control, and an uncertainty-stratified attribution analysis.

The answer is qualified rather than unconditional, and it is stated that way on purpose. On accuracy the model is competitive on dissolution and weight uniformity, behind on tensile strength, and clearly behind tuned gradient boosting on hardness, the only difference reaching significance. On multi-task structure the answer is positive: one shared encoder matches or improves on four separate networks across all four targets, significantly so on dissolution, using 3.73 times fewer parameters. On trustworthiness the answer is positive with a caveat: intervals are near nominal on three of four attributes, error and uncertainty both respond sharply to a genuine change of regime while staying stable on an unseen product inside the trained range, and the sensors the model relies on reorder measurably as confidence falls ( $\tau = 0.822$ ,  $p = 0.0004$ ). Weight uniformity is poorly calibrated and is reported as such.

The hardness shortfall was not left unexplained. Error across held-out product codes scales almost linearly with distance from the centre of the training distribution ( $r = 0.941$ ), which identifies the cause as the concentration of the high-hardness regime in a few product codes rather than a failure of the architecture.

The contribution is therefore not a uniformly more accurate predictor, but a demonstration that a single trajectory-native model can predict four correlated attributes without feature engineering, at a fraction of the parameters of separate models, while providing calibrated uncertainty that responds to real distribution shift and explanations conditioned on its own confidence. None of the work reviewed in Section 2 provides that combination.

### 7.1 Future work

Four directions follow from the limitations above. **Repeated or nested cross-validation** with a hyperparameter search would give the significance tests usable power. **Targeted data collection at the extremes** follows from the hardness relationship: the most valuable batches are those furthest from the centre of the distribution. **Hybrid uncertainty estimation** using a small ensemble of evidential models may recover the

ensemble’s robustness under shift while keeping single-pass efficiency. **Cross-site validation** on a second press is necessary before any claim of general applicability. The input-representation ablation (Nie et al., 2023) remains open.

The demo app shows what a deployable form would look like when batches from on unfamiliar regime are visibly flagged by the models own widened uncertainty. It is a support tool and at the same time a research artifact as well, and not a replacement for the Lab testing, Nothing in this study supports using it as one as well.

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